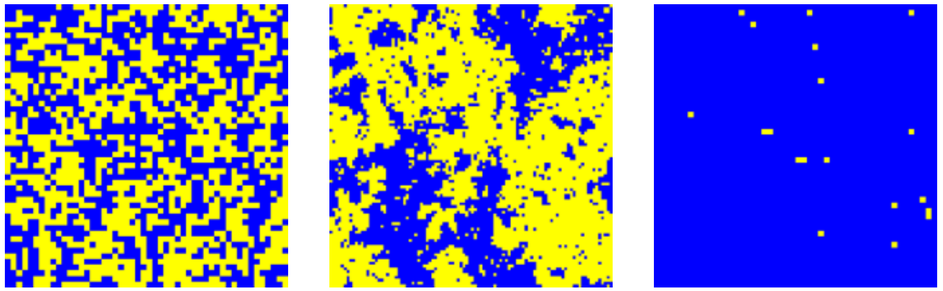
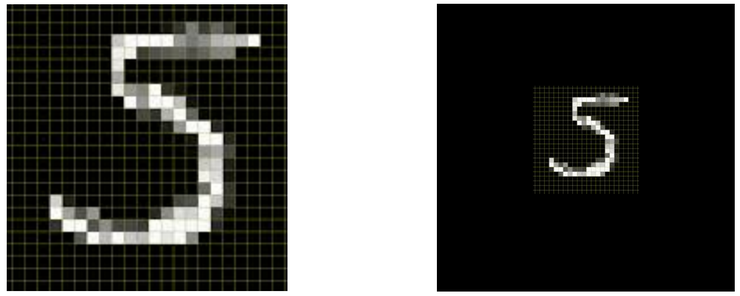
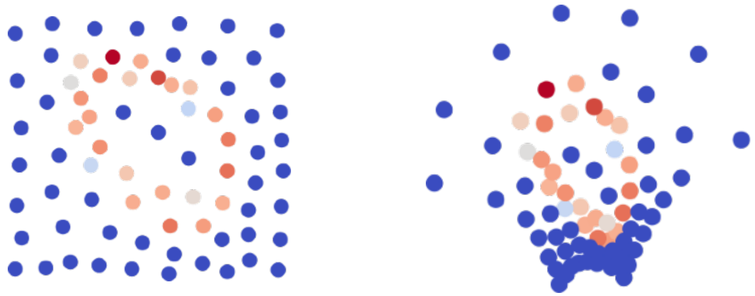
Uses of scale and conformal invariance

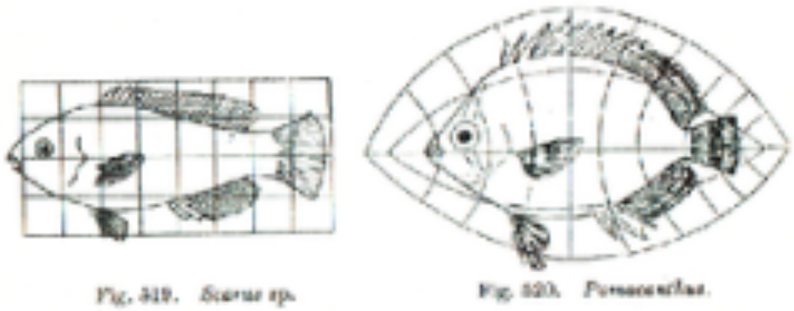
- Conformal invariance plays an important role in physics. Applications range from phase transitions — e.g. the Curie point in magnets — to quantum gravity (this will actually be relevant later!)

- In computer and human vision there is a useful notion of scale (and possibly conformal) invariance (e.g. Lei et. al 2023).





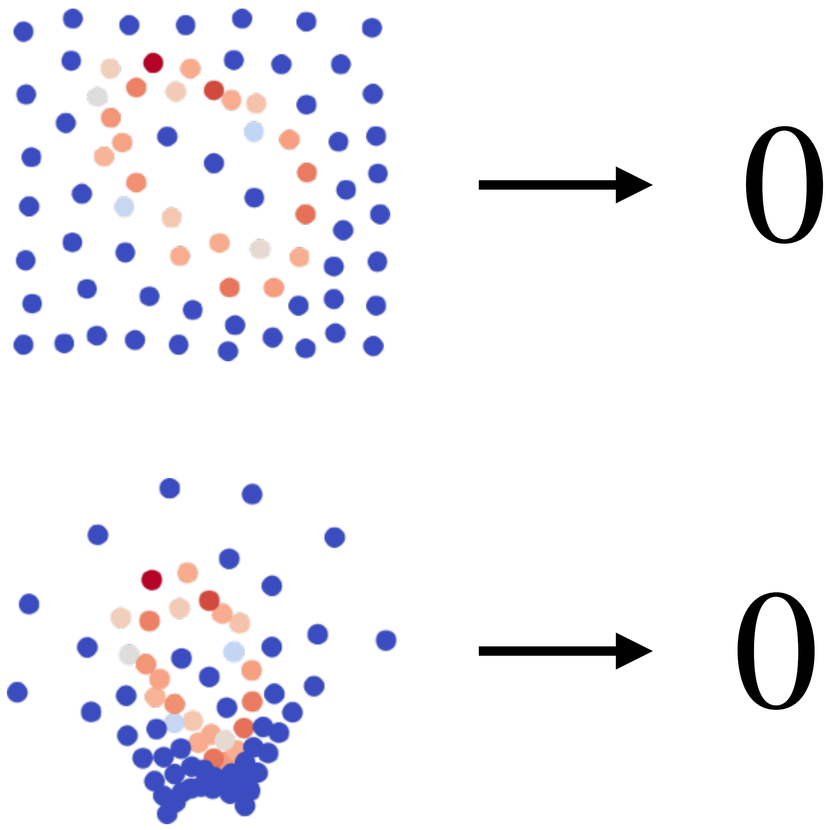




Of Growth and Form, Thompson 1917

Conformally equivariant neural networks

- We want to build a neural network that takes in a cloud of points and is **invariant** under conformal transformations: so we get exactly the same answer if we conformally transform each point.

- What is an **efficient way to process the point cloud** in a way that transforms nicely under conformal symmetry?

- Standard methods are difficult (conformal group is **non-compact** and **high-dimensional**)